

Physiology Of Neurons

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Learning Outcomes:

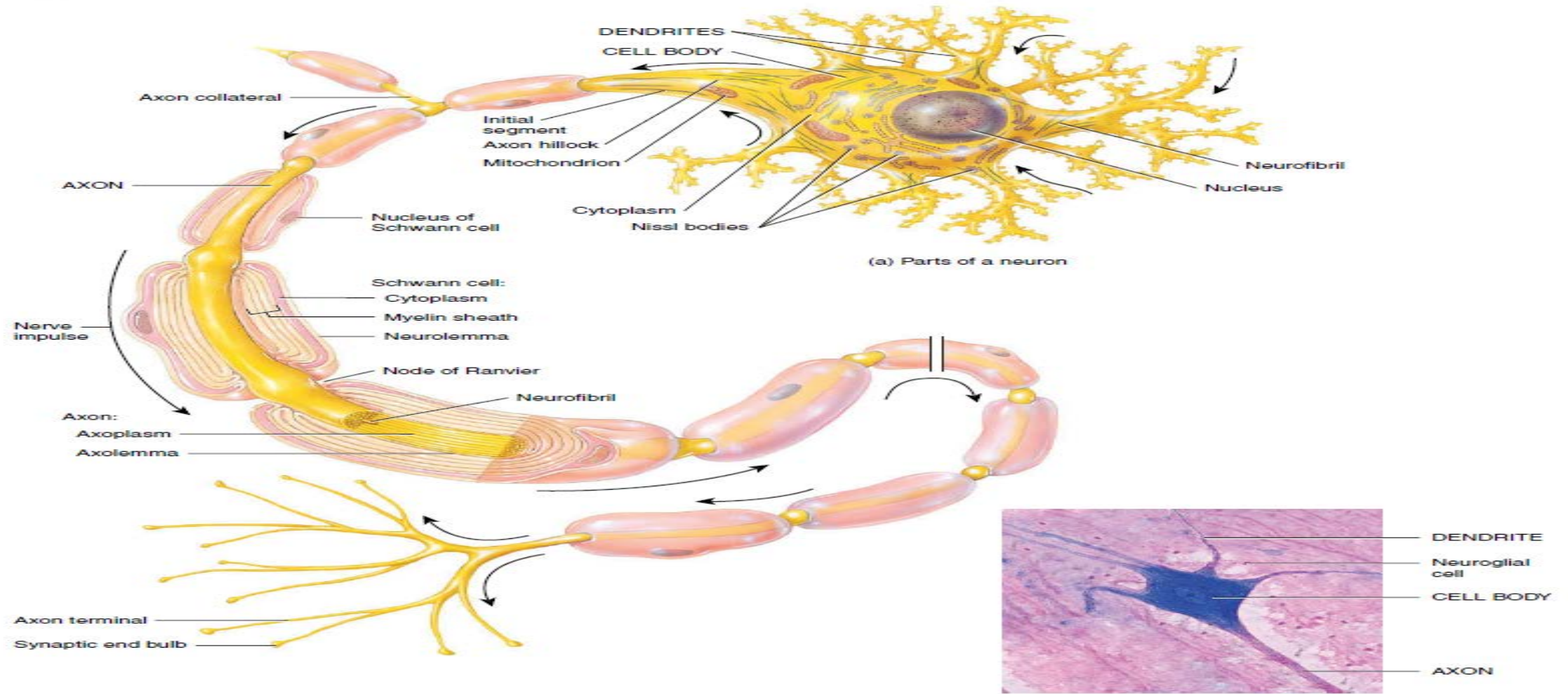
- Classify Neurons Functionally.
- Summarize the functions of Neuroglia.
- Define Myelination.
- Explain the processes of electrical signaling in neurons.

What are Neurons?

- Neurons (nerve cells) possess electrical excitability, the ability to respond to a stimulus and convert it into an action potential.
- A stimulus is any change in the environment that is strong enough to initiate an action potential.

Parts of a Neuron

Most neurons have three parts: (1) a cell body, (2) dendrites, and (3) an axon.



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- In most neurons, nerve impulses arise at an area called the **trigger zone**.
 - A nerve fiber is a general term for any neuronal process (extension) that emerges from the cell body of a neuron. Most neurons have two kinds of processes: multiple dendrites and a single axon.

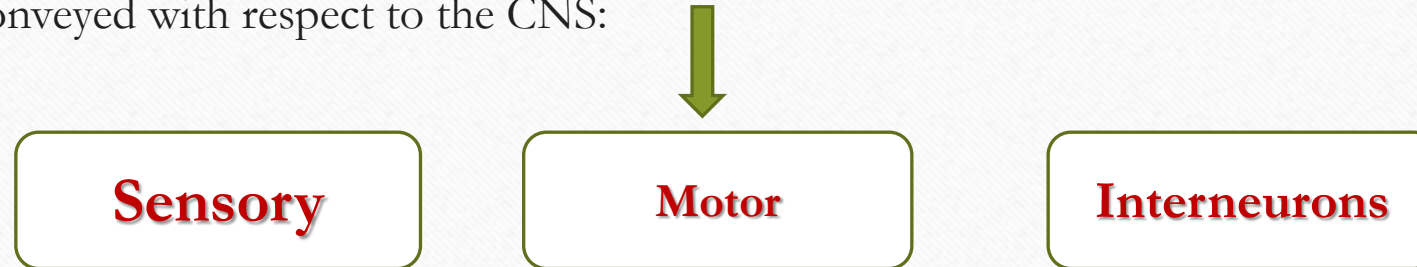
What is Synapse?

What are Neurotransmitters?

- The site of communication between two neurons or between a neuron and an effector cell is a Synapse.
- synaptic end bulbs store a chemical called a **neurotransmitter**.
- A neurotransmitter is a molecule released from a synaptic vesicle in synaptic end bulb that excites or inhibits another neuron, muscle fiber, or gland cell.

Classification of Neurons

- Functionally, neurons are classified according to the direction in which the nerve impulse (action potential) is conveyed with respect to the CNS:



- 1. Sensory or afferent neurons:** either contain sensory receptors at their distal ends (dendrites) or are located just after sensory receptors that are separate cells. Once a stimulus activates a sensory receptor, the sensory neuron forms an action potential in its axon and the action potential is conveyed into the CNS through cranial or spinal nerves.

Classification of Neurons

- 2. **Motor or efferent neurons:** convey action potentials away from the CNS to effectors (muscles and glands) in the periphery (PNS) through cranial or spinal nerves.
- 3. **Interneurons or association neurons:** are mainly located within the CNS between sensory and motor neurons. Interneurons integrate (process) incoming sensory information from sensory neurons and then elicit a motor response by activating the appropriate motor neurons.

Neuroglia

- Neuroglia or glia make up about half the volume of the CNS.
- Their name derives from the idea of early histologists that they were the “glue” that held nervous tissue together.
- Neuroglia are not merely passive bystanders but rather actively participate in the activities of nervous tissue.
- In contrast to neurons, glia do not generate or propagate action potentials.
- In cases of injury or disease, neuroglia multiply to fill in the spaces formerly occupied by neurons.

Functions Of Neuroglia

Neuroglia of the CNS include astrocytes, oligodendrocytes, microglial cells, and ependymal cells:

- ☐ **Astrocytes:** several functions ex. support neurons and play a role in learning and memory.
- ☐ **Oligodendrocytes:** form the myelin sheath around CNS axons.
- ☐ **Microglia:** function as phagocytes.
- ☐ **Ependymal cells:** produce and assist in the circulation of cerebrospinal fluid.

Functions Of Neuroglia

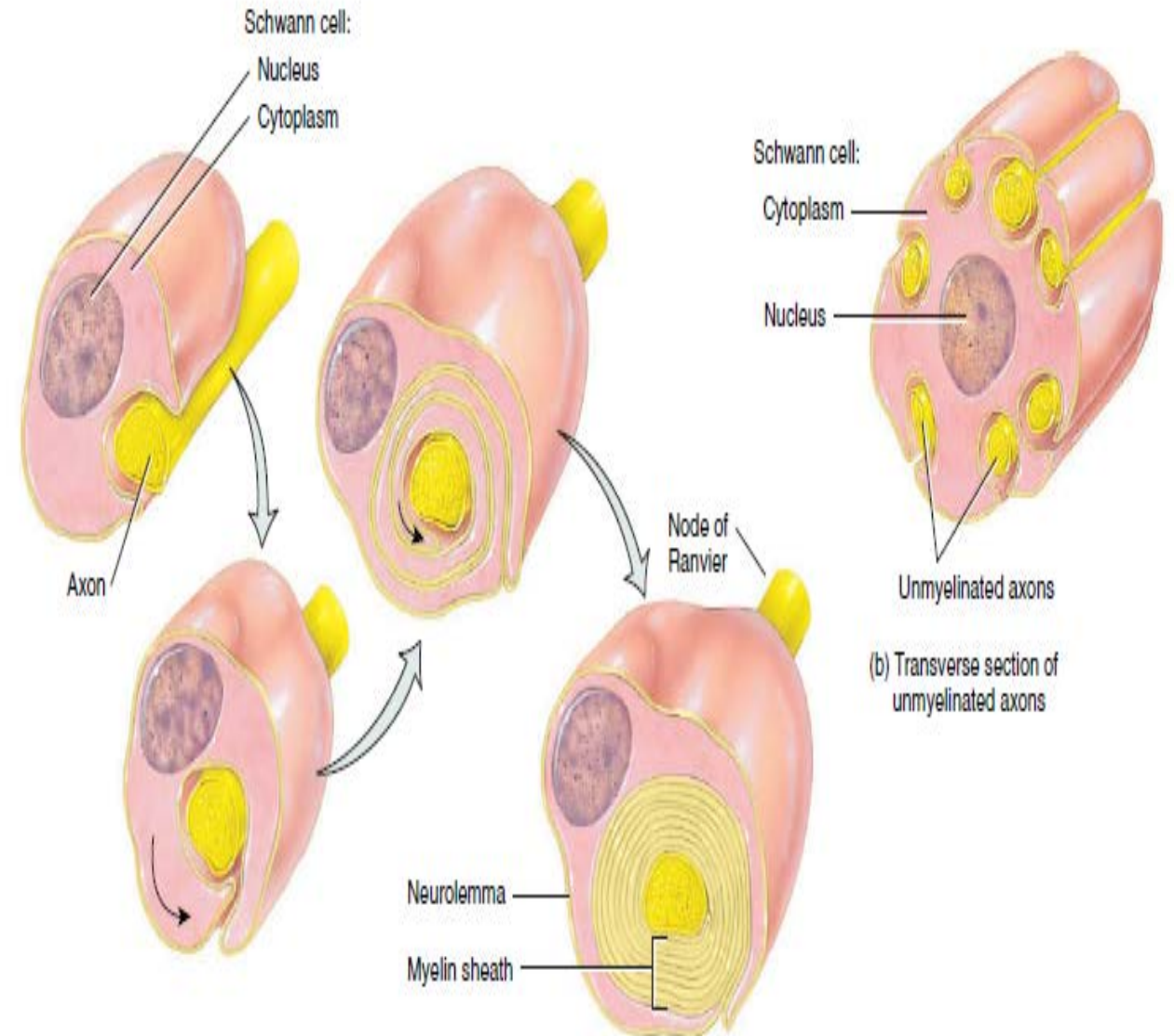
Neuroglia of the PNS are Schwann cells and satellite cells:

- ❑ **Schwann Cells:** form the myelin sheath around axons of the PNS.
- ❑ **Satellite Cells:** regulate the exchanges of materials between neuronal cell bodies and interstitial fluid.

What is Myelination?

- The myelin sheath is a multilayered lipid and protein covering around some axons that insulates them and increases the speed of nerve impulse conduction. Such axons are said to be **myelinated**.

 Axons surrounded by a myelin sheath produced either by Schwann cells in the PNS or by oligodendrocytes in the CNS are said to be myelinated.



(a) Transverse sections of stages in the formation of a myelin sheath

Important Terms:

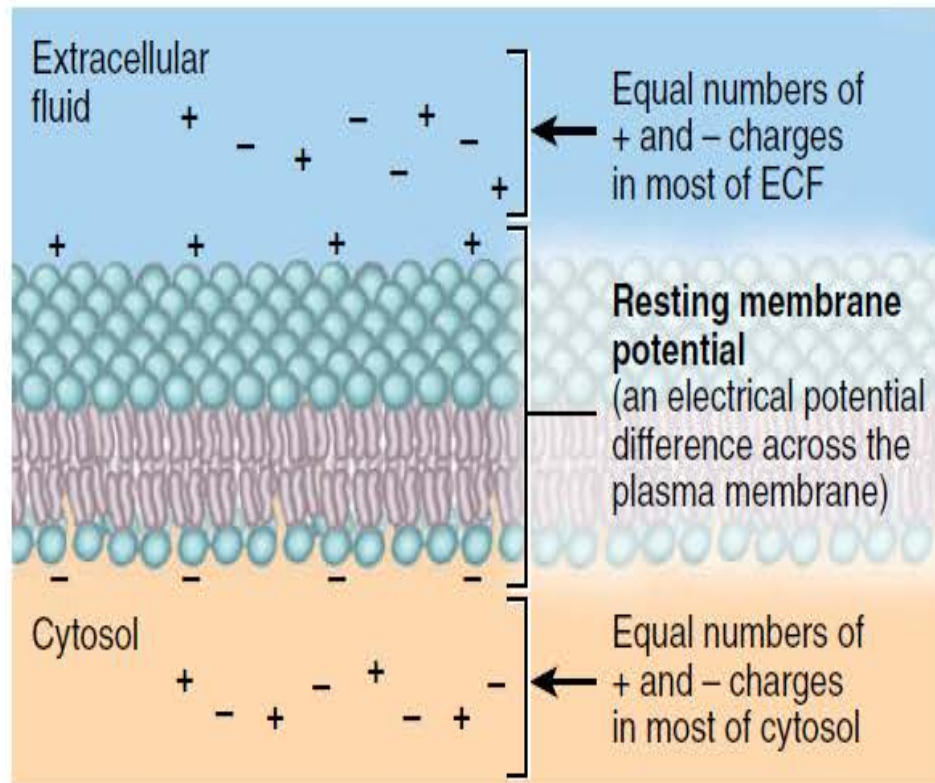
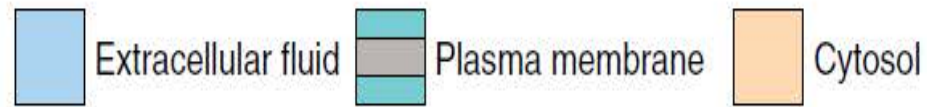
- **Neurolemma:** The outer nucleated cytoplasmic layer of the Schwann cell, which encloses the myelin sheath, is the neurolemma, neurolemma is found only around axons in the PNS. When an axon is injured, the neurolemma aids regeneration by forming a regeneration tube that guides and stimulates regrowth of the axon.
- **Ganglion:** a cluster of neuronal cell bodies located in the PNS.
- **Nucleus:** is a cluster of neuronal cell bodies located in the CNS.
- **Nerve:** is a bundle of axons that is located in the PNS. Cranial nerves connect the brain to the periphery, whereas spinal nerves connect the spinal cord to the periphery.

Electrical Signals In Neurons

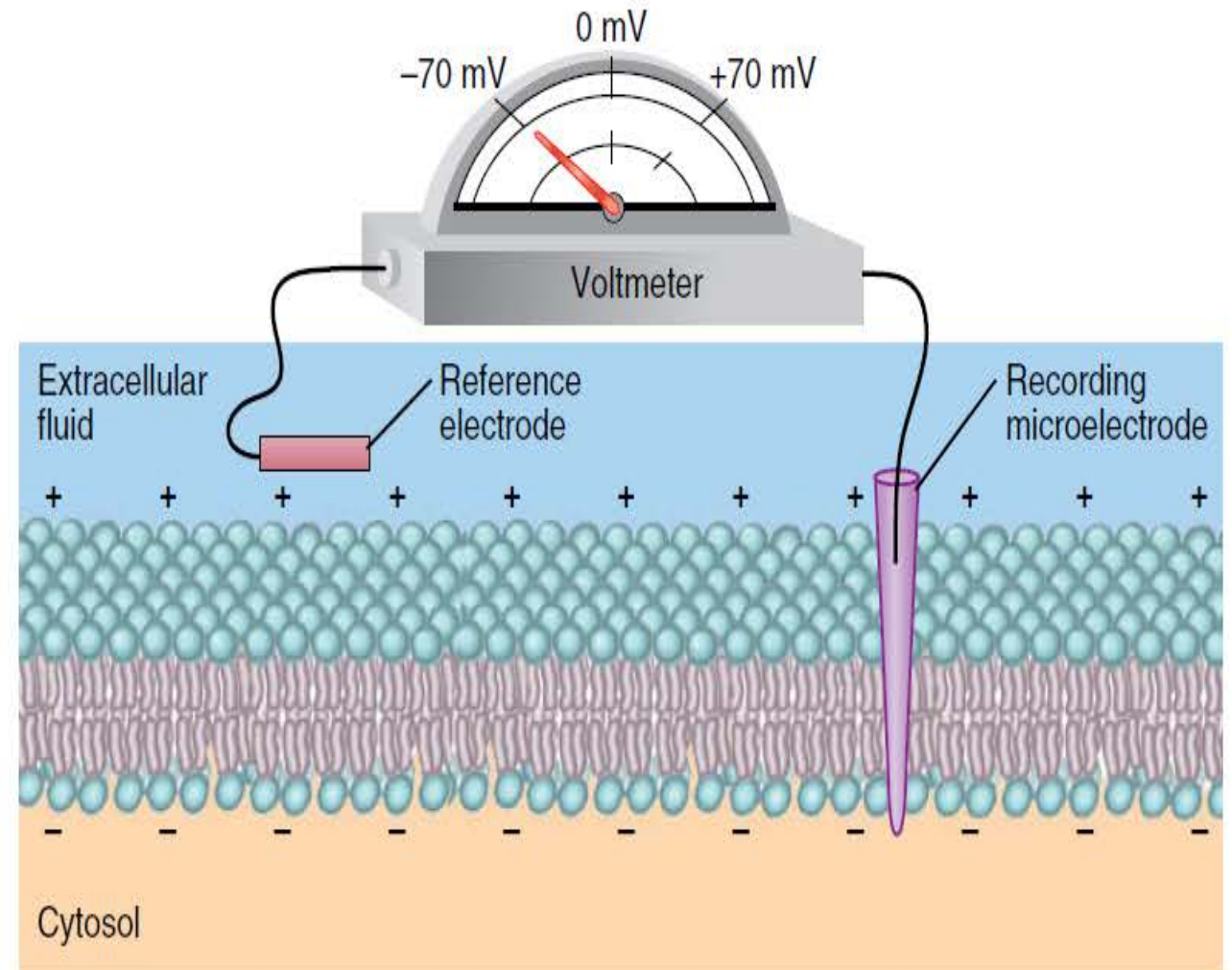
- Like muscle fibers, neurons are electrically excitable.
- They communicate with one another using two types of electrical signals:
 - (1) **Graded potentials**: are used for short- distance communication only.
 - (2) **Action potentials**: allow communication over long distances within the body.

What Is Resting Membrane Potential?

- The resting membrane potential exists because of a small buildup of negative ions in the cytosol along the inside of the membrane, and an equal buildup of positive ions in the extracellular fluid (ECF) along the outside surface of the membrane.
- Such a separation of positive and negative electrical charges is a form of potential energy.
- The buildup of charge occurs only very close to the membrane. The cytosol or extracellular fluid elsewhere in the cell contains equal numbers of positive and negative charges and is electrically neutral.



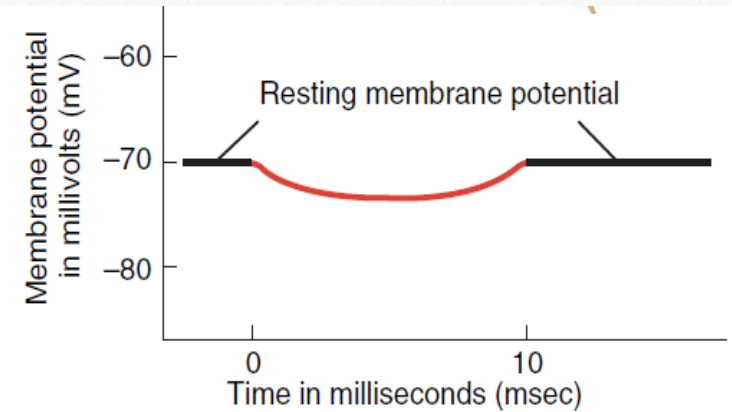
(a) Distribution of charges that produce the resting membrane potential of a neuron



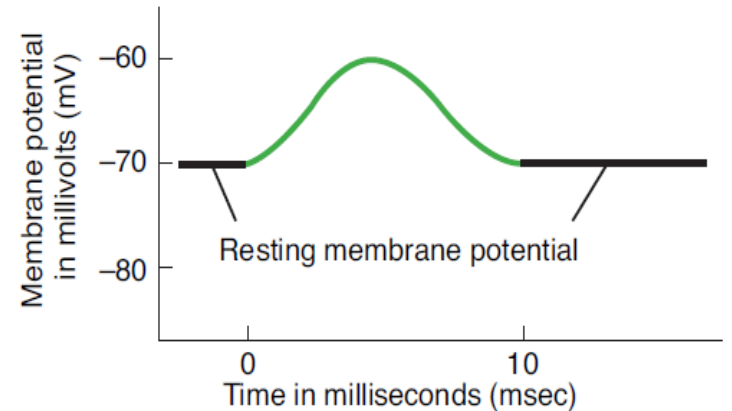
(b) Measurement of the resting membrane potential of a neuron

What is Graded Potential?

- A graded potential is a small deviation from the resting membrane potential that makes the membrane either more polarized (inside more negative) or less polarized (inside less negative).
- When the response makes the membrane more polarized (inside more negative), it is termed a hyperpolarizing graded potential. When the response makes the membrane less polarized (inside less negative), it is termed a depolarizing graded potential.
- A graded potential occurs when a stimulus causes mechanically-gated or ligand-gated channels to open or close in an excitable cell's plasma membrane.



(a) Hyperpolarizing graded potential



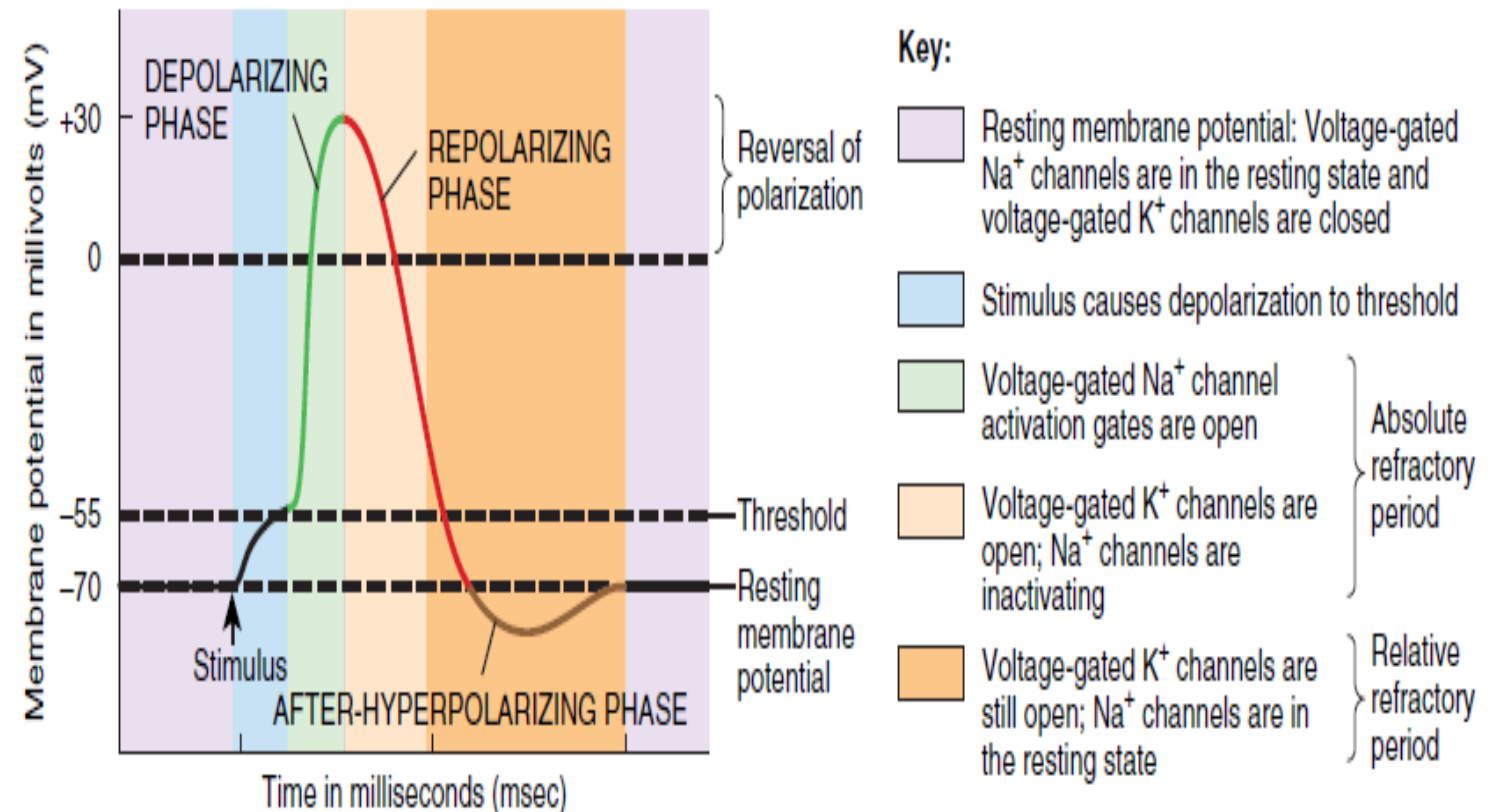
(b) Depolarizing graded potential

What is Action Potential?

- An action potential (nerve impulse) is an electrical signal that propagates (travels) along the surface of the membrane of a neuron. It begins and travels due to the movement of ions (such as sodium and potassium) between interstitial fluid and the inside of a neuron through specific ion channels.
- Once begun, a nerve impulse travels rapidly and at a constant strength.

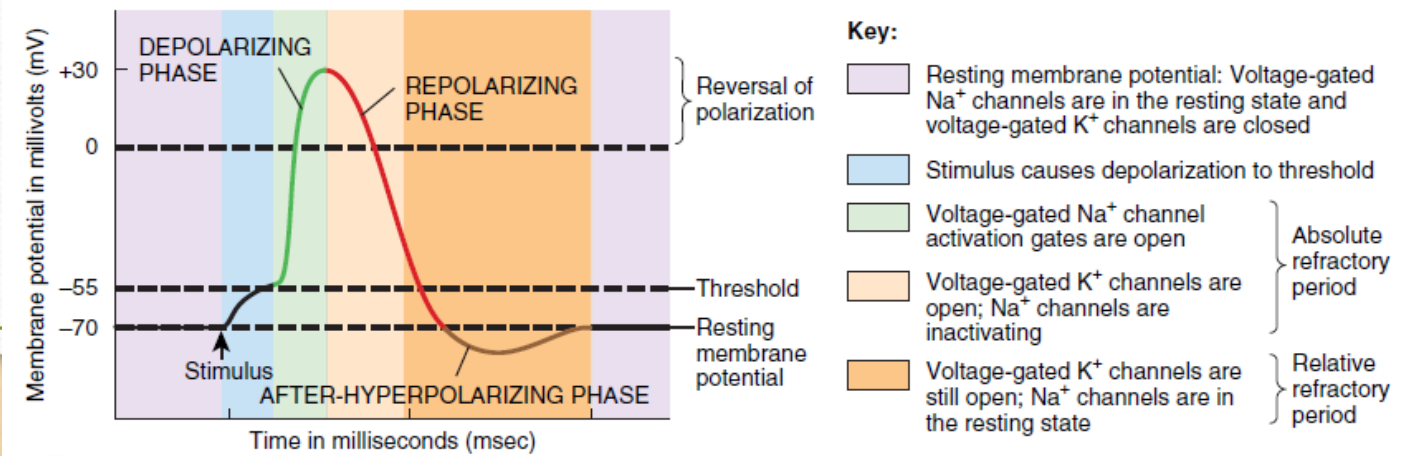
Generation of Action Potentials

- An action potential (AP) decrease and reverse the membrane potential and then eventually restore it to the resting state.
- An action potential has two main phases: a depolarizing phase and a repolarizing phase.



Generation of Action Potentials

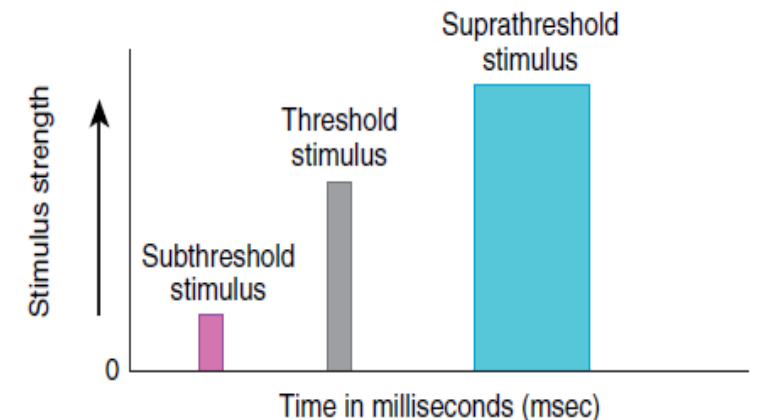
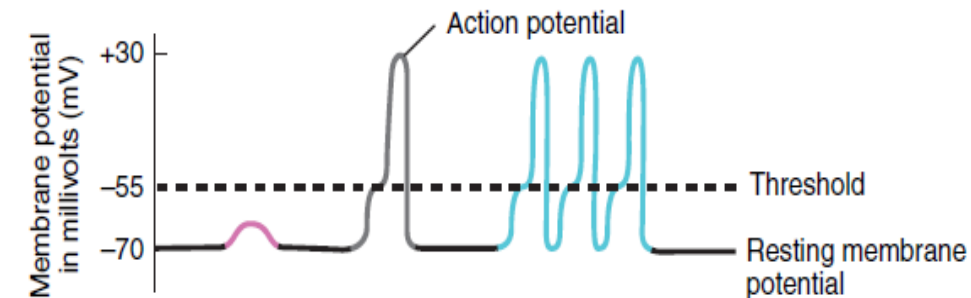
- During the depolarizing phase, the negative membrane potential becomes less negative, reaches zero, and then becomes positive. During the repolarizing phase, the membrane potential is restored to the resting state of -70 mV.
- Following the repolarizing phase there may be an after-hyperpolarizing phase, during which the membrane potential temporarily becomes more negative than the resting level.



Generation of Action Potentials

- Ion channels open and then close during an action potential. These channels are present mainly in the axon plasma membrane and axon terminals.
- The first channels that open, the voltage-gated Na^+ channels, allow Na^+ to rush into the cell, which causes the depolarizing phase.

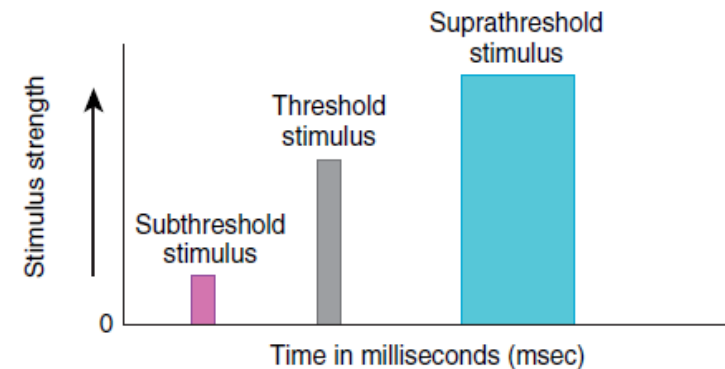
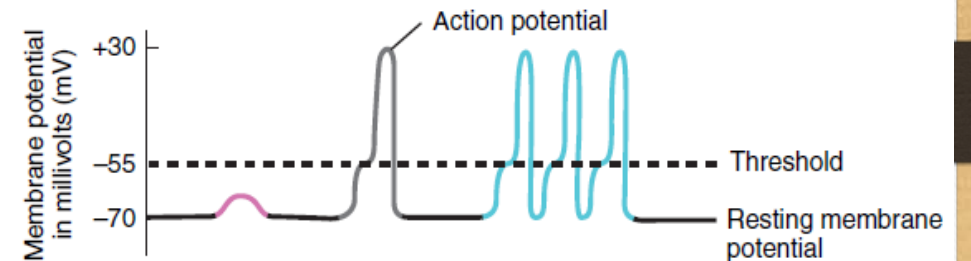
An action potential will only occur once the membrane potential reaches threshold.



Generation of Action Potentials

- Then voltage gated K^+ channels open, allowing K^+ to flow out, which produces the repolarizing phase. The after-hyperpolarizing phase occurs when the voltage-gated K channels remain open after the repolarizing phase end.
- An action potential occurs in the membrane of the axon of a neuron when depolarization reaches a certain level termed the threshold (about -55 mV in many neurons).

An action potential will only occur once the membrane potential reaches threshold.





*THANK
YOU*